SECTION A

1) (B) $CH_3C^*HClCH_2Br$
2) (B) H_2/Ni
3) (A) $[Cr(H_2O)_5Cl]Cl_2 \cdot H_2O$
4) (C) Proteins
5) (D) 4
6) (C) CH_3NH_2
7) (A) Shows large negative deviation from Rault's law
8) (C) Zero order reaction
9) (D) Relative Lowering of Vapour pressure
10) (A) They are chemically reactive
11) (D) $\frac{R}{2K}$
12) (B) Sn is oxidised and voltage of the cell is 0.91 V

13) (a) Both Assertion (A) and Reason (R) are true and Reason is the correct explanation of Assertion

14) (a) Both (A) and (R) are true and (R) is the correct explanation of (A)

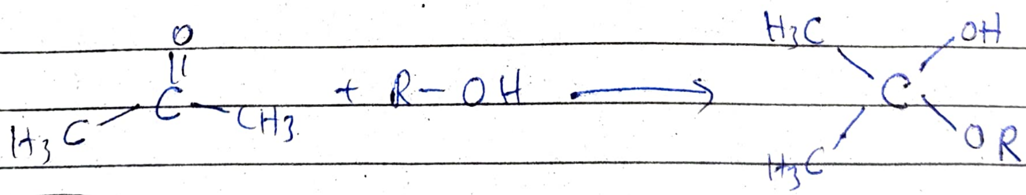
15) Assertion is false but reason is true

16) Both (A) and (R) are true and (R) is correct explanation of (A)

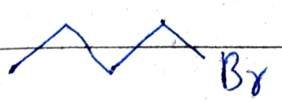
SECTION - B

Q17) (i) When a pressure greater than osmotic pressure is applied reverse osmosis start.

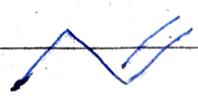
(ii) When acetone is added to pure alcohol Ketal is formed



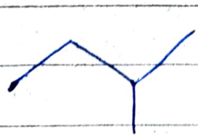
Q18) (A)



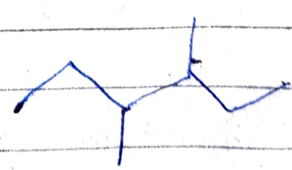
(B)



(C)



(D)



PAGE NO.:
DATE: / /

Q19) (i) Oligosaccharides are the carbohydrates which on hydrolysis gives 2 to 10 units of monosaccharides

(ii) Invert sugars are the sugars which have their configuration inverted from normal sugars

Q20) (i) $Cr_2O_7^{2-} + 14H^+ + 6Fe^{2+} \longrightarrow 6Fe^{3+} + 2Cr^{3+} + 7H_2O$

(ii) $Cr_2O_7^{2-} + 14H^+ + 3Sn^{2+} \longrightarrow 3Sn^{4+} + 2Cr^{3+} + 7H_2O$

Q21) (a) Bond angle ~~Fe~~ (C—O—H) is slightly less than the tetrahedral angle because the higher extend of repulsion between the lone pairs (e^- pairs) of oxygen

(b) Due to the effect of resonance in phenol there exists a partial double bond character in C–OH of phenol, which decreases it's bond length than in CH_3OH

SECTION-C

Q22) (i) Cr^{2+}, because it's e^- config is $t_{2g}^3 e_g^1$ which on oxidation converts into a stable $t_{2g}^3 e_g^0$ configuration

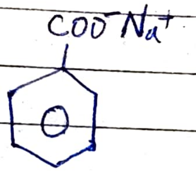
(ii) Fe^{3+}, due to ~~Fe^{3+}/Fe^{2+}~~ $E^0_{Fe^{3+}/Fe^{2+}} > 0$

(iii) Cu^{2+}, because its configuration is $3d^9 4s^0$ which on reduction (addition of e^-) converts to a stable fully filled $3d^{10}$ configuration

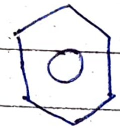
Q23) (a) A → $CH_3-\overset{\overset{O}{\|}}{C}H$ B → $H_3C-CH=N-OH$

(b) A → $CH_3-\overset{\overset{O}{\|}}{C}-OH$ B → $CH_3-\overset{\overset{O}{\|}}{C}-Cl$

(C) A →



B →



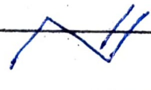
Q24)

$H_2C=CH_2 \xrightarrow{H^+/H_2SO_4} H_2\overset{+}{C}-CH_3$

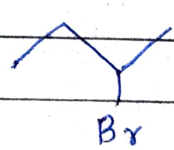
$H_2\ddot{O}$

$H^+ + H_2\underset{OH}{C}-CH_3 \longleftarrow H_2\overset{\overset{+}{O}H_2}{C}-CH_3$

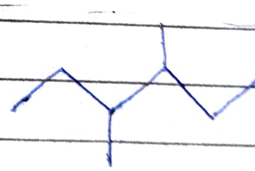
Q25) A →



B →

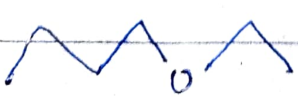


C →



D → $CH_3CH_2CH_2^-(MgBr)^+$ E → $CH_3CH_2CH_3$

F →



PAGE NO.:
DATE: / /

Q 26) (i) Aniline does not undergo Friedel-crafts reactions because in Aniline the 'N' of the NH_2 group is a stronger nucleophile than the para, ortho position of benzene ring hence desired product is not formed

(ii) Aromatic primary amines cannot be prepared by Gabriel's phthalimide synthesis because aryl halides do not undergo nucleophilic substitution

(iii) Aliphatic amines are stronger bases than ammonia because of the +I effect of the (R) group

in H_2N-R

Q 27) We know $\Delta T_f = K_f (i)(m)$

$\Rightarrow$ for $AlCl_3 \longrightarrow Al^{+} + 3Cl^{-}$

$1-\alpha \qquad \alpha \qquad 3\alpha$

$\boxed{i = 1+3\alpha}$

$0.068 = 1.86(1+3\alpha)(0.01)$

$\frac{6.8}{1.86} = 1+3\alpha$

$\alpha = 0.883$

% dissociation $= 100\alpha = 88.3\%$ approx

PAGE NO
DATE / /

Q28) As they are connected in series current will be same = 2A

$2g\ Cu = \frac{2}{63.5}$ mol Cu

for converting 1 mol Cu^{2+} to Cu we require 1 F electricity

for $\frac{2}{63.5}$ mol $\rightarrow$ $\frac{2}{63.5}$ F is required

now same electricity is passed in Zn cell
let mol of Zn formed be x

$x\cancel{F} = \frac{2}{63.5}\cancel{F}$ $\rightarrow$ $x = \frac{2}{63.5}$

mass $= x \times 65 = \frac{2}{63.5} \times 65 = \boxed{2.04\ g}$

SECTION-D

Q29) (i) $\frac{1}{2} + \frac{3}{2} = 2$

(ii) Rate of Reaction means the ~~time~~ rate at which one stiatiometric coeff changes in concentration

PAGE NO

DATE / /

(iii) for a 1^{st} order reaction

$$A = A_0 e^{-Kt}$$

$$3 = 5 e^{-(10^{-3})t}$$

$$\Rightarrow \frac{3}{5} = e^{-Kt} \quad \Rightarrow \log\left(\frac{5}{3}\right) \times 2.303 = 10^{-3} t$$

$$\Rightarrow (.2219)\ 2.303 \times 10^{3} = t$$

$$.50666 \times 10^{3} = t$$

$$\boxed{t = 506 \text{ sec approx}}$$

Q30) (i) glycosidic linkage is the linkage between to monosaccharide or higher to form Oligo or poly saccharides
while
peptide linkages are linkages between amino acids to form proteins

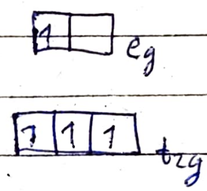
(ii) Essential amino acids are those which cannot be synthesized in our body
e.g. Leucine, Isoleucine, Lysine, Methionine etc

(iii) When a protein is given a significant temperature, pH or change in it's physical and chemical conditions, it loses it's biological activity ~~known~~ this whole process is known as denaturation of proteins

During denaturation, secondary and teritary structures loose their ~~then~~ activity while primary structures remain intact.

SECTION-E

Q31) (a)(i) if $\Delta_o < P$ then e^- will go in e_g orbital than pairing up in t_{2g}



so e^- config $\rightarrow$ $t_{2g}^3 e_g^1$

(ii) dsp^2, diamagnetic

(iii) $[Co(en)_2Cl_2]^+$

(b) (i) $[Ni(H_2O)_6]Cl_2$

(ii) Hexaaqua nickel (II) chloride

Q 32) (a) $Zn_{(s)} + Cu^{2+}_{(aq)} \longrightarrow Zn^{2+}_{(aq)} + Cu_{(s)}$

net $E^o_{cell} = 1.1\ V$

$\Delta G^o = -nFE^o_{cell}$ $\boxed{n = 2}$

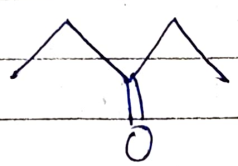
$\Delta G^o = -2 \times 96500 \times 1.1$
$= -212300\ J/mol$
$= \boxed{-212.3\ KJ/mol}$

(b) Two advantages of fuel cells are as follows

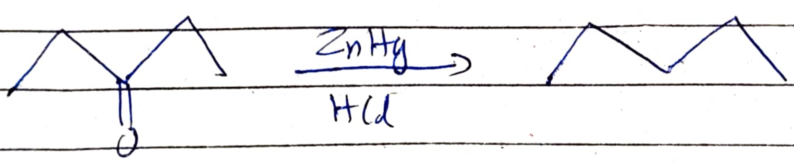
1) They have high efficiency

2) ~~They~~ We donot have to make space for storing electrolytes ~~use~~ as we can pass gas when we require output leading low standing loss

Q 33) (a) A →

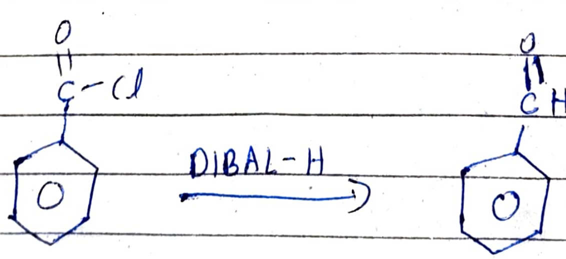


pentan-3-one
or
3-pentanone



(b) (i) $CH_3-CH_2-COOH \xrightarrow[\text{(ii) } H_2O]{\text{(i) } Br_2/\text{Red P}} CH_3-\underset{\displaystyle Br}{\underset{|}{CH}}-COOH$

(ii)



(c) We can use Fehling's test to distinguish between benzaldehyde and acetaldehyde as benzaldehyde do not give +ve Fehling's test.